Azure Cosmos DB SQL API Assignment

Create

```
Database: ev_network
Container: charging_stations
Partition Key: /city
```

Data


```
{
  "id": "CS001",
  "station_name": "VoltHub Central",
  "city": "Hyderabad",
  "zone": "Central",
  "station_type": "Fast",
  "connector_type": "CCS2",
  "capacity_kw": 120,
  "daily_sessions": 46,
  "revenue": 18400,
  "uptime_percent": 98.5,
  "status": "Active"
}
```

```
{
  "id": "CS002",
  "station_name": "GreenCharge East",
  "city": "Hyderabad",
  "zone": "East",
  "station_type": "Standard",
  "connector_type": "Type2",
  "capacity_kw": 60,
  "daily_sessions": 31,
  "revenue": 9300,
  "uptime_percent": 96.2,
```

```
  "status": "Active"
}

{
  "id": "CS003",
  "station_name": "RapidVolt TechPark",
  "city": "Bengaluru",
  "zone": "East",
  "station_type": "Fast",
  "connector_type": "CCS2",
  "capacity_kw": 150,
  "daily_sessions": 58,
  "revenue": 24700,
  "uptime_percent": 99.1,
  "status": "Active"
}

{
  "id": "CS004",
  "station_name": "UrbanPlug North",
  "city": "Bengaluru",
  "zone": "North",
  "station_type": "Standard",
  "connector_type": "Type2",
  "capacity_kw": 50,
  "daily_sessions": 26,
  "revenue": 7800,
  "uptime_percent": 94.8,
  "status": "Maintenance"
}

{
  "id": "CS005",
  "station_name": "ChargeBay Marina",
  "city": "Chennai",
  "zone": "South",
  "station_type": "Fast",
  "connector_type": "CCS2",
  "capacity_kw": 100,
  "daily_sessions": 43,
  "revenue": 16500,
  "uptime_percent": 97.4,
```

```
  "status": "Active"
}

{
  "id": "CS006",
  "station_name": "EcoPlug OMR",
  "city": "Chennai",
  "zone": "East",
  "station_type": "Standard",
  "connector_type": "Type2",
  "capacity_kw": 40,
  "daily_sessions": 22,
  "revenue": 6400,
  "uptime_percent": 95.1,
  "status": "Active"
}

{
  "id": "CS007",
  "station_name": "PowerGrid Koregaon",
  "city": "Pune",
  "zone": "East",
  "station_type": "Fast",
  "connector_type": "CCS2",
  "capacity_kw": 120,
  "daily_sessions": 39,
  "revenue": 15100,
  "uptime_percent": 98.0,
  "status": "Active"
}

{
  "id": "CS008",
  "station_name": "PlugPoint Hinjewadi",
  "city": "Pune",
  "zone": "West",
  "station_type": "Standard",
  "connector_type": "Type2",
  "capacity_kw": 60,
  "daily_sessions": 34,
  "revenue": 10200,
  "uptime_percent": 96.9,
```

```
  "status": "Active"
}

{
  "id": "CS009",
  "station_name": "MetroCharge Andheri",
  "city": "Mumbai",
  "zone": "West",
  "station_type": "Fast",
  "connector_type": "CCS2",
  "capacity_kw": 180,
  "daily_sessions": 67,
  "revenue": 29600,
  "uptime_percent": 99.3,
  "status": "Active"
}

{
  "id": "CS010",
  "station_name": "Harbour EV Point",
  "city": "Mumbai",
  "zone": "South",
  "station_type": "Standard",
  "connector_type": "Type2",
  "capacity_kw": 50,
  "daily_sessions": 29,
  "revenue": 8700,
  "uptime_percent": 93.7,
  "status": "Maintenance"
}

{
  "id": "CS011",
  "station_name": "CapitalCharge Connaught",
  "city": "Delhi",
  "zone": "Central",
  "station_type": "Fast",
  "connector_type": "CCS2",
  "capacity_kw": 150,
  "daily_sessions": 61,
  "revenue": 26100,
  "uptime_percent": 98.7,
```

```
  "status": "Active"
}

{
  "id": "CS012",
  "station_name": "NorthGrid Rohini",
  "city": "Delhi",
  "zone": "North",
  "station_type": "Standard",
  "connector_type": "Type2",
  "capacity_kw": 60,
  "daily_sessions": 33,
  "revenue": 9900,
  "uptime_percent": 95.8,
  "status": "Active"
}

{
  "id": "CS013",
  "station_name": "PinkCity Charge",
  "city": "Jaipur",
  "zone": "Central",
  "station_type": "Fast",
  "connector_type": "CCS2",
  "capacity_kw": 100,
  "daily_sessions": 35,
  "revenue": 13300,
  "uptime_percent": 97.1,
  "status": "Active"
}

{
  "id": "CS014",
  "station_name": "DesertPlug West",
  "city": "Jaipur",
  "zone": "West",
  "station_type": "Standard",
  "connector_type": "Type2",
  "capacity_kw": 40,
  "daily_sessions": 18,
  "revenue": 5100,
  "uptime_percent": 92.5,
```

```
  "status": "Maintenance"
}

{
  "id": "CS015",
  "station_name": "LakeCity EV Hub",
  "city": "Bhopal",
  "zone": "Central",
  "station_type": "Fast",
  "connector_type": "CCS2",
  "capacity_kw": 90,
  "daily_sessions": 32,
  "revenue": 11800,
  "uptime_percent": 96.7,
  "status": "Active"
}

{
  "id": "CS016",
  "station_name": "SmartPlug Arera",
  "city": "Bhopal",
  "zone": "South",
  "station_type": "Standard",
  "connector_type": "Type2",
  "capacity_kw": 45,
  "daily_sessions": 21,
  "revenue": 5900,
  "uptime_percent": 94.3,
  "status": "Active"
}

{
  "id": "CS017",
  "station_name": "CoastalCharge Panampilly",
  "city": "Kochi",
  "zone": "South",
  "station_type": "Fast",
  "connector_type": "CCS2",
  "capacity_kw": 110,
  "daily_sessions": 38,
  "revenue": 14600,
  "uptime_percent": 98.2,
```

```
  "status": "Active"
}
```

```
{
  "id": "CS018",
  "station_name": "Backwater Plug",
  "city": "Kochi",
  "zone": "North",
  "station_type": "Standard",
  "connector_type": "Type2",
  "capacity_kw": 50,
  "daily_sessions": 24,
  "revenue": 7200,
  "uptime_percent": 95.4,
  "status": "Active"
}
```


Section A — Basic SELECT

1. Display all documents in the container.
2. Display only `station_name`, `city`, and `status`.
3. Display `station_name`, `capacity_kw`, `daily_sessions`, and `revenue`.
4. Display `id` and `station_name` for every station.

Section B — WHERE Filters

5. Display all stations in Hyderabad.
6. Display all stations in Mumbai.
7. Display all stations whose status is `Maintenance`.
8. Display all `Fast` charging stations.
9. Display all stations whose connector type is `CCS2`.

Section C — Numeric Filters

10. Display stations whose capacity is greater than `100`.
11. Display stations whose revenue is greater than `15000`.

12. Display stations having more than `40` daily sessions.
13. Display stations whose uptime is below `96`.
14. Display stations whose capacity is between `50` and `120`.
15. Display stations whose revenue is below `10000`.

Section D — AND Conditions

16. Display stations that are `Fast` and `Active`.
17. Display Hyderabad stations whose revenue is above `10000`.
18. Display stations whose capacity is greater than `100` and daily sessions are greater than `40`.
19. Display Active stations whose uptime is greater than `97`.
20. Display Standard stations whose revenue is below `10000`.

Section E — OR Conditions

21. Display stations located in Hyderabad or Bengaluru.
22. Display stations whose zone is North or South.
23. Display stations that are Fast or have revenue above `20000`.
24. Display stations located in Mumbai, Delhi, or Chennai.

Section F — ORDER BY

25. Display all stations ordered by revenue from lowest to highest.
26. Display all stations ordered by revenue from highest to lowest.
27. Display stations ordered by daily sessions from highest to lowest.
28. Display stations ordered by uptime percentage from lowest to highest.
29. Display stations ordered first by city and then by revenue.

Section G — COUNT

30. Count the total number of charging stations.
31. Count the number of Active stations.
32. Count the number of Fast stations.
33. Count the number of stations in Pune.

34. Count the number of stations whose revenue is greater than `15000`.

Section H — GROUP BY

35. Count the number of stations in each city.
36. Count the number of stations by `station_type`.
37. Count the number of stations by `status`.
38. Find the total revenue for each city.
39. Find the average revenue for each station type.
40. Find the average uptime percentage for each city.
41. Find the total number of daily sessions for each city.
42. Find the average charging capacity for each station type.

Section I — GROUP BY with Filters

43. Find total revenue by city using only Active stations.
44. Count Fast stations in each city.
45. Find average revenue by city for stations whose revenue is above `8000`.
46. Find total daily sessions grouped by station type.

Section J — Combined Queries

47. Find the top five stations based on revenue.
48. Find the top three stations based on daily sessions.
49. Display Active Fast stations ordered by revenue from highest to lowest.
50. Display stations whose revenue is greater than `12000` and uptime is greater than `97`, ordered by revenue from highest to lowest.
51. Display Hyderabad or Bengaluru stations having more than `30` daily sessions.
52. Display Maintenance stations ordered by uptime percentage from lowest to highest.
53. Display `station_name`, `city`, `revenue`, and `uptime_percent` for stations whose revenue is greater than `15000`.
54. Display Fast stations having a capacity of at least `120`, ordered by capacity from highest to lowest.

Section K — Business Questions

55. Write a query to determine which city has the highest total revenue.
56. Write a query to determine which station has the highest revenue.
57. Write a query to determine which station has the lowest uptime percentage.
58. Write a query to determine which station type has the highest average revenue.
59. Write a query to determine which city has the highest total number of daily charging sessions.
60. Write a query to determine how many stations are currently under maintenance.